

BALIKESIR, Turkey

WMO: 171500

Lat: 39.617N

Lon: 27.917E

Elev: 102

StdP: 100.11

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	-3.8	-8.0	1.9	-3.5	-6.6	2.2	-1.8	12.5	4.2	10.9	6.5	0.7	350	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	35.9	20.6	34.0	20.4	32.7	20.3	22.9	31.0	22.2	30.2	21.4	29.4	4.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.1	26.3	19.6	14.5	26.0	18.8	13.8	25.7	68.2	31.0	65.5	30.2	63.0	29.3	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.9	8.9	DB	-8.7	39.6	3.7	1.8	-11.4	40.9	-13.6	41.9	-15.7	42.9	-18.4	44.2	
			WB	-9.1	24.0	3.7	1.0	-11.7	24.7	-13.9	25.3	-16.0	25.8	-18.7	26.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	4.8	6.3	8.8	12.9	17.9	22.4	25.1	25.2	21.1	16.1	10.4	6.3	
	DBStd	8.02	4.22	4.25	3.76	3.50	3.03	2.68	2.03	2.18	2.82	3.54	4.15	4.40	
	HDD10.0	542	168	115	68	12	0	0	0	0	0	3	44	132	
	HDD18.3	1973	420	336	294	167	45	2	0	0	8	87	239	374	
	CDD10.0	2301	7	12	32	98	244	372	468	472	334	191	56	16	
	CDD18.3	691	0	0	0	2	31	124	209	214	92	16	1	0	
	CDH23.3	8026	0	0	7	81	497	1507	2447	2347	953	176	10	1	
CDH26.7	3531	0	0	1	14	153	657	1191	1106	366	43	1	0		
Wind	WSAvg	3.0	2.6	3.0	3.0	2.5	2.5	3.3	4.3	4.5	3.2	2.8	2.1	2.5	
Precipitation	PrecAvg	577	82	66	59	50	41	25	9	8	25	43	78	97	
	PrecMax	996	274	200	139	148	102	96	53	52	120	202	173	300	
	PrecMin	354	2	4	13	5	4	0	0	0	0	0	3	0	
	PrecStd	132	57	40	31	28	25	25	13	13	28	37	40	61	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	20.1	24.0	28.8	33.0	37.2	39.0	37.9	35.8	30.8	24.9	19.5	
		MCWB	11.9	12.6	14.2	17.0	18.5	20.4	21.0	20.6	19.2	18.2	16.1	13.3	
	2%	DB	15.9	17.7	21.0	25.8	30.2	34.1	35.9	35.1	32.9	27.6	21.8	17.1	
		MCWB	11.2	11.8	12.9	16.2	18.2	19.9	20.7	20.8	19.1	17.4	14.7	12.4	
	5%	DB	13.9	15.7	19.0	23.2	28.2	32.2	34.0	33.5	30.3	25.1	19.3	15.1	
		MCWB	10.1	10.8	12.2	15.0	17.5	19.6	20.6	21.0	18.7	16.6	13.8	11.5	
	10%	DB	11.8	13.5	16.8	21.0	26.1	30.4	32.1	32.0	28.7	22.9	17.1	13.0	
		MCWB	8.9	9.8	11.3	14.1	16.9	19.2	20.5	20.8	18.3	16.0	12.9	10.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	13.6	15.0	18.2	20.2	22.7	23.5	24.1	22.0	19.7	17.2	13.9
			MCDB	16.8	18.6	22.4	26.0	28.8	32.3	32.1	32.3	30.1	26.0	22.8	18.1
2%		WB	11.3	12.2	13.6	16.7	19.2	21.5	22.7	23.2	20.8	18.5	15.5	12.7	
		MCDB	15.2	16.9	19.6	24.3	27.8	30.1	31.3	30.8	28.1	24.6	20.3	16.4	
5%		WB	10.2	11.0	12.5	15.5	18.3	20.7	22.0	22.5	19.9	17.6	14.3	11.7	
		MCDB	13.4	15.0	18.0	22.3	26.3	29.1	30.5	29.9	27.1	23.3	18.6	14.9	
10%		WB	9.0	10.0	11.3	14.3	17.4	19.9	21.3	21.7	19.1	16.7	13.1	10.5	
		MCDB	11.6	13.4	16.2	20.3	24.8	28.3	29.7	29.0	26.5	21.7	16.8	12.9	
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.4	11.4	13.0	14.4	14.7	13.9	13.4	14.0	11.9	10.9	7.9	
		MCDBR	11.9	13.7	16.8	17.8	18.7	18.2	18.1	16.5	17.8	16.3	14.2	10.8	
		MCWBR	8.2	8.8	9.9	9.9	8.6	6.9	6.7	5.8	7.2	8.1	8.7	7.2	
	5% WB	MCDBR	11.0	12.5	15.0	16.4	16.2	15.3	14.3	13.5	14.2	13.3	12.8	10.0	
		MCWBR	7.9	8.2	9.1	9.4	8.1	6.3	5.7	5.1	6.2	7.0	8.2	6.8	
Clear-Sky Solar Irradiance	taub		0.363	0.397	0.438	0.482	0.452	0.424	0.422	0.430	0.419	0.408	0.376	0.364	
	taud		2.321	2.197	2.090	1.998	2.152	2.277	2.287	2.260	2.277	2.281	2.318	2.327	
	Ebn at Noon		785	804	809	799	833	856	854	837	821	784	761	749	
	Edh at Noon		92	119	147	172	150	133	131	131	121	109	92	86	
All-Sky Solar Radiation	RadAvg		1.76	2.44	3.75	5.02	6.29	7.18	7.60	6.66	5.07	3.41	2.27	1.62	
	RadStd		0.24	0.32	0.36	0.57	0.45	0.54	0.23	0.32	0.35	0.30	0.28	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		+0.57	N/A	N/A	N/A	N/A	N/A	-55	-137	+119	+63
Regional (18 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-107	+112	+58

Nomenclature: See separate page